

Sanjay J K

Software Engineer — AI Systems Developer
+91-8122394864 | j.k.sanjay2006@gmail.com | Portfolio | LinkedIn | GitHub

SUMMARY

Computer Science student at VIT (CGPA 9.34) building scalable full-stack software systems, from event-driven architectures to production-grade web applications. Experienced in API design, databases, and system architecture, with practical experience in AI/ML, including LLM pipelines and Retrieval-Augmented Generation (RAG). Holder of two granted patents in applied AI.

SKILLS

Systems & Backend: Event-Driven Architecture, REST APIs, WebSockets, OAuth2, Circuit Breakers, Asynchronous Job Processing, Data Structures & Algorithms, Docker, Redis, PostgreSQL, MongoDB, SQL, ChromaDB, Neo4j

AI / ML: LLM Pipelines, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, NLP

Languages: Python, TypeScript, JavaScript, C++, Java, C

Frameworks & Web: React, Next.js, FastAPI, Node.js, Express.js, TailwindCSS, Zustand

Tools: Git, GitHub, Pytest, LangChain, LangGraph, Langfuse, Ollama, Docker Compose

Certifications: IBM Generative AI

EDUCATION

Vellore Institute of Technology (VIT) — B.Tech Computer Science and Engineering

2023 – 2027

CGPA: 9.34

PROJECTS

LLM Knowledge Assistant (Production RAG System)

Python, LangChain, ChromaDB

- Engineered a **hybrid retrieval pipeline** (chunking, MiniLM embeddings, BM25 + dense vector search via Reciprocal Rank Fusion, cross-encoder re-ranking) that lifted precision by **31%** and reached a perfect **100% Hit Rate@5** on a golden evaluation set.
- Built a citation-confidence gate that declines low-relevance queries, validated via ablation studies (Recall@K, MRR, NDCG) with full Langfuse tracing across each retrieval stage.

InterviewAI — Real-Time Voice AI Interviewer

FastAPI, React, WebSockets, Ollama

- Built a real-time voice AI interviewer (Deepgram ASR, Ollama LLMs, ElevenLabs TTS) achieving **sub-1.2s end-to-end latency** via an event-driven, full-duplex WebSocket streaming architecture.
- Engineered a **zero-lag interruption handler** that cancels in-flight LLM/TTS generation and flushes client audio buffers mid-response, enabling natural conversational turn-taking.

NextUp — AI-Powered Placement Intelligence Platform

Next.js, FastAPI, PostgreSQL, Redis

- Architected a **zero-knowledge security system** (client-side AES-256-GCM encryption, peppered HMAC blind-index matching) across **112 REST API routes**, matching shortlists without the server ever accessing plaintext student data.
- Built an **LLM extraction pipeline with deterministic grounding gates** that re-verify every model-extracted field against source text, eliminating hallucinated dates and eligibility criteria from a small model on free-tier infra.
- Diagnosed and fixed an N+1 query bottleneck on a single 512MB production server, cutting a core endpoint's response time from **30s+ to 2.3s**, and added circuit-breaker failover across a multi-provider AI gateway for zero-downtime degradation.

Prefilter AI — Query Understanding Middleware

Python, Qwen 0.8B LoRA, FastAPI

- Built a **fine-tuned SLM query parser** (Qwen 0.8B LoRA adapters) that extracts structured search filters from natural language across 10 domains, never hallucinating values not explicitly mentioned in the query.
- Engineered backend translators that compile parsed queries into **SQL, MongoDB, ChromaDB, Elasticsearch** filters without re-parsing, plus a **ConflictDetector** module flagging feasibility violations (e.g., budget mismatches) with automatic constraint relaxation.

PATENTS & PUBLICATIONS

- Published (2026, Indian Patent Office):** Personalized AI System for Color Vision Deficiency — an adaptive recommendation engine using perceptual gamut modeling to assist colorblind users.
- Published (2026, Indian Patent Office):** Pan-Chronological Vision-Language Transformer for Cross-Era Tamil Script Decipherment — a VLM system for reading historical Tamil inscriptions across centuries of script drift.
- SET2026 Conference (Accepted, Presenting 2026):** Co-authored a physics-augmented ML digital twin for thermosyphon heat pipe thermal management — 7 ML modules (classification, regression, uncertainty quantification, anomaly detection); **97.9% F1** steady-state detection, **R²=0.91** surrogate accuracy on unseen geometries.

EXPERIENCE

International Research Scholar — UKIERI-SPARC Program, Heriot-Watt University

Jun 2025

- Selected as **1 of 9 students nationally** for an international hydrogen-energy research exchange; applied iterative, data-driven experimentation (cyclic & linear sweep voltammetry) to catalyst validation, drawing on the same analysis-and-abstraction habits used in debugging and systems work.
- Co-presented a research-to-startup roadmap (technical findings, budget, and scalability plan) to a faculty panel, translating experimental data into a structured business case.

Frontend Developer Intern — Valsco Technology (Remote)

Jun 2024 – Sep 2024

- Scaled **NoteSwap** to **200+ active student users** by architecting a component-based React frontend with reusable UI modules and REST API integration, cutting page load time by **20%** through lazy loading and render optimization.
- Improved cross-device usability by designing a responsive component library adopted across the full desktop and mobile interface.

AWARDS & RECOGNITION

- GATE CSE 2026: AIR 1603** (Score: **670**).
- Judges' Choice Award – Sankalp Innovation Challenge 2026 (MNNIT Allahabad):** Built **JanVedha AI**, a civic governance platform with grievance analytics, explainable AI prioritization (LightGBM/SHAP), and a Gemini-powered decision assistant.
- Finalist – Google Cloud Agent AI Hackathon 2025 & India Innovates 2026;** Top 3 – DeepShiva LLM Hackathon.
- Board Member (Design Lead), AEE VIT;** Senior Core Member, CodeChef VIT & TAM-VIT.